16. STRUCTURING OF THE VENOUS SYSTEM 2 : THE VITELLINE & UMBILICAL SYSTEM

DURATION : 08:22 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth exploration of the vitelline and umbilical systems, focusing on the development of the liver from mesodermal and endodermal tissues. It details the different stages of formation of the hepatic vascular network, including the vitelline and umbilical veins, as well as their complex interactions. Key concepts such as cephalisation, cardialisation, and the two currents of the hepatic portal system are explained, highlighting the importance of this network for blood drainage and overall metabolic health. The emotional impacts related to digestive organs are also addressed, thus enriching the biodynamic understanding of the human body.

17 - STRUCTURING THE VENOUS SYSTEM 2: THE VITELLINE & UMBILICAL SYSTEM

THE VITELLINE SYSTEM AND THE LIVER

The **vitelline system** is the second major network we will discuss, in connection with the digestive system. The **liver** is the main receptacle of this system.

It is essential to remember that the liver results from the combination of two tissues: a **mesodermal tissue** and an **endodermal tissue**.

My endodermal tube has an upper, middle, and lower intestine. The liver emerges at the junction between the upper and middle intestine, initially developing as a "field of aspiration," a subtle description of the digestive tissue.

The **20 vitelline veins**, mesodermal structures, are at the origin of liver formation.

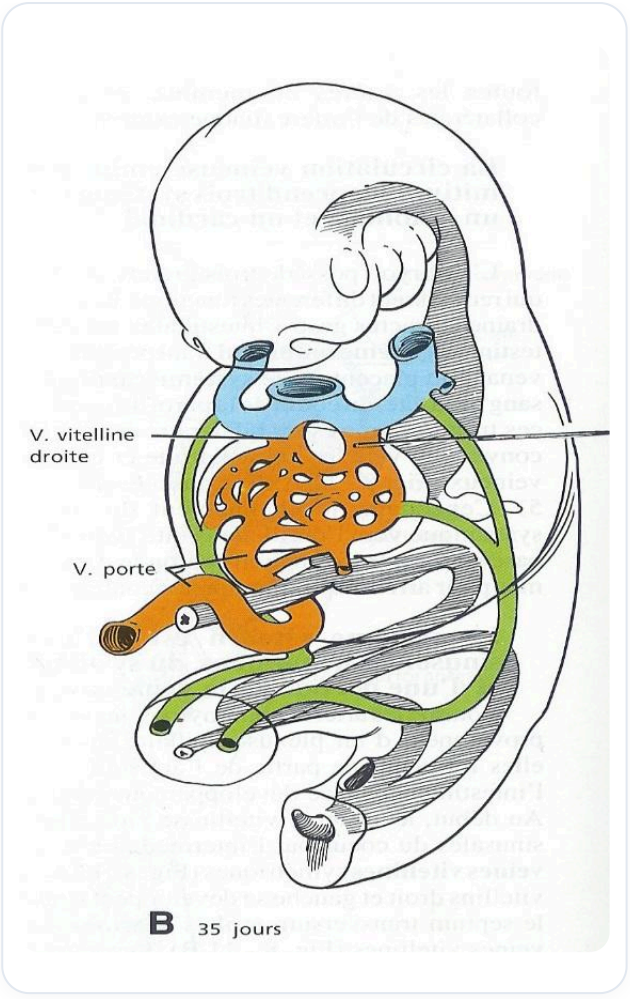


FIG. — Embryo hepatic vascular system

LIVER DEVELOPMENT SEQUENCE

Cephalisation leads to **cardialisation**, which causes **diaphragmatisation**. This process results in hepatic congestion at the mesodermal level, through vitelline integration.

The movement of cephalisation leads to cardialisation, then diaphragmatisation, leading to a phase of congestion below the diaphragm. This area of congestion corresponds to the mesodermal development of the liver. Initially, the liver receives no umbilical information, but is informed by disassimilation products from the brain, lower trunk, and vitelline vesicle.

Its first impulse is therefore a reception of **toxicity**. The liver, as an organ, has the primary function of receiving and transforming this toxicity, before having a regenerative impulse later.

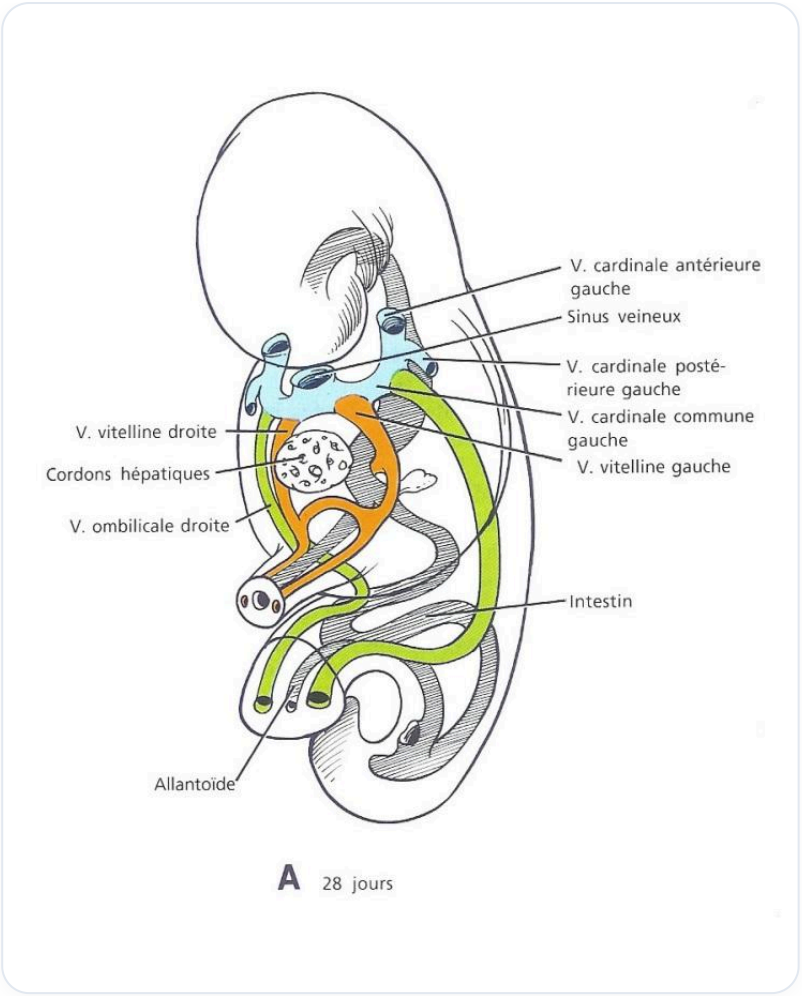


FIG. — Embryonic circulation 28 days

FORMATION OF THE HEPATIC VASCULAR NETWORK

We have the **right vitelline vein** and the **left vitelline vein**, with a small anastomosis already formed between the two. In parallel, the hepatic cord originating from the endoderm develops.

Between these two vitelline veins, an entire network forms, congests, and organises. This network will eventually form the venous duct and, later, structures such as the **gastric vein** and the **gastro-epiploic vein**. An important area will develop: the **portal vein**.

At another level, the **umbilical vein** develops, part of which will be integrated while another part will disappear. This network of vitelline systems meets to form a specific vascular network, acting as a drainage system.

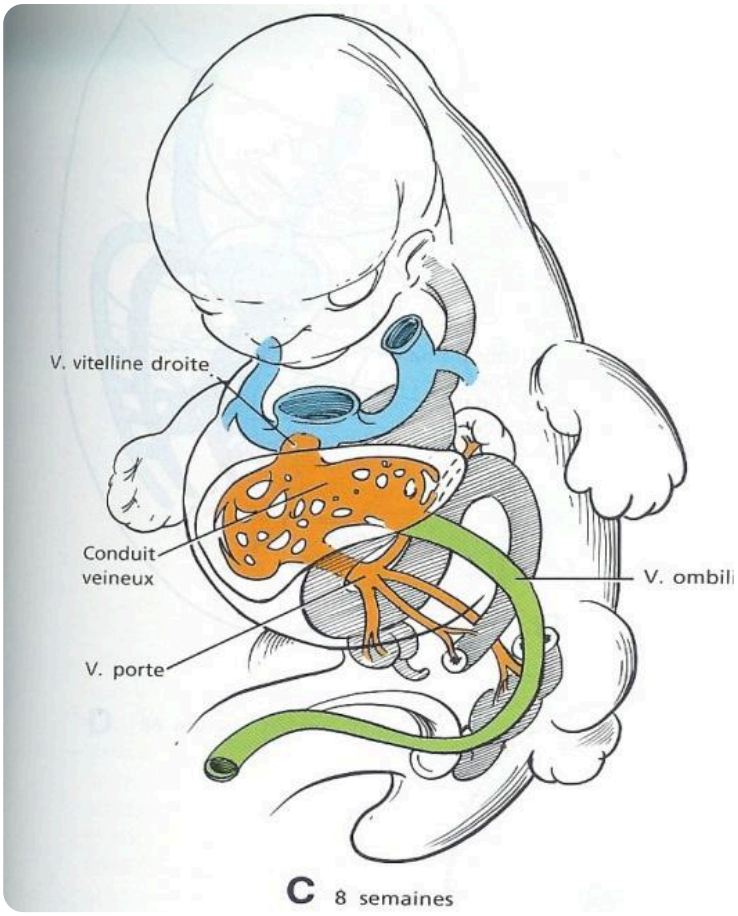


FIG. — Embryonic circulation 8 weeks

SEGMENTATION OF THE VITELLINE VEIN

Development takes place under the **septum transversum**. The vitelline vein, located under this septum, can be divided into three sections:

- **Suprahepatic section:** It will eventually form the hepatic duct, then the hepatic vein. The right vitelline vein becomes the hepatic vein, the region of flow towards the heart.
- **Hepatic section.**
- **Infrahepatic section.**

Another derivative of this vitelline part is the upper part of the **inferior vena cava**.

REGRESSIONS AND MODIFICATIONS OF DRAINAGE

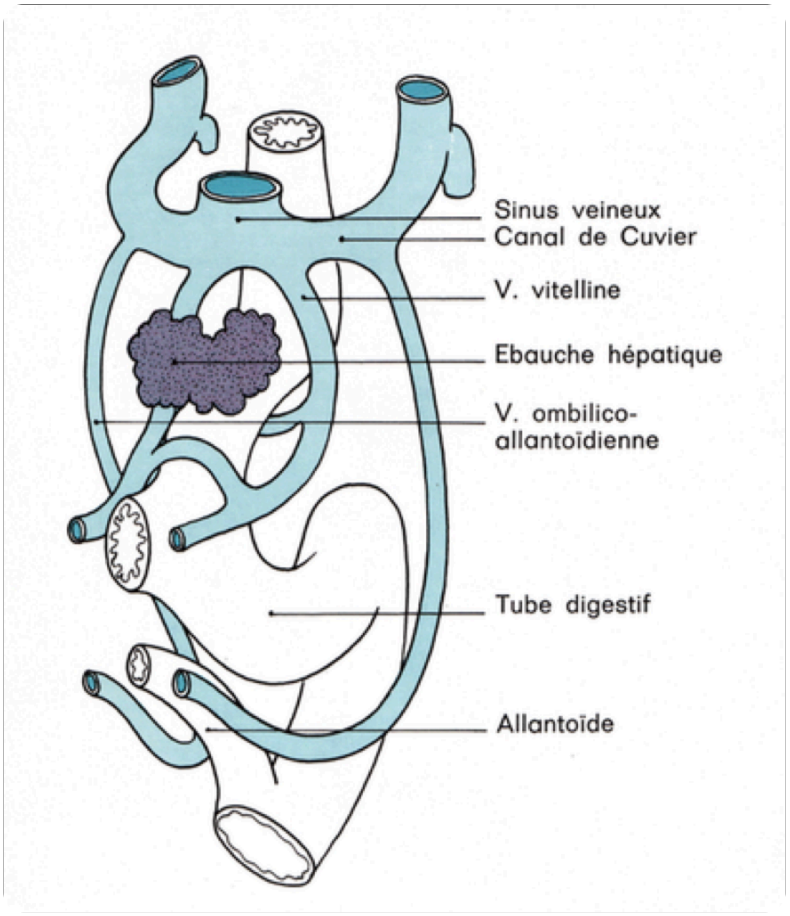


FIG. — Embryonic venous circulation

Regressions occur. The left umbilical vein regresses, but part of the right umbilical vein persists, forming a link for the **hepatic sinusoids**.

The umbilical vitelline vein is here. The left umbilical vein forms a link, then gradually disappears in its upper part. The **ductus venosus** will be formed with the portal trunk.

In the infrahepatic region, the left vitelline vein regresses, but develops an anastomosis with the right vitelline vein, leading to a change in blood drainage. Blood passes from the left side of the abdominal viscera to the right vitelline vein, creating a developing transverse network. This leads to a **switch** from the left side to the right side at the venous level, promoting growth on the right.

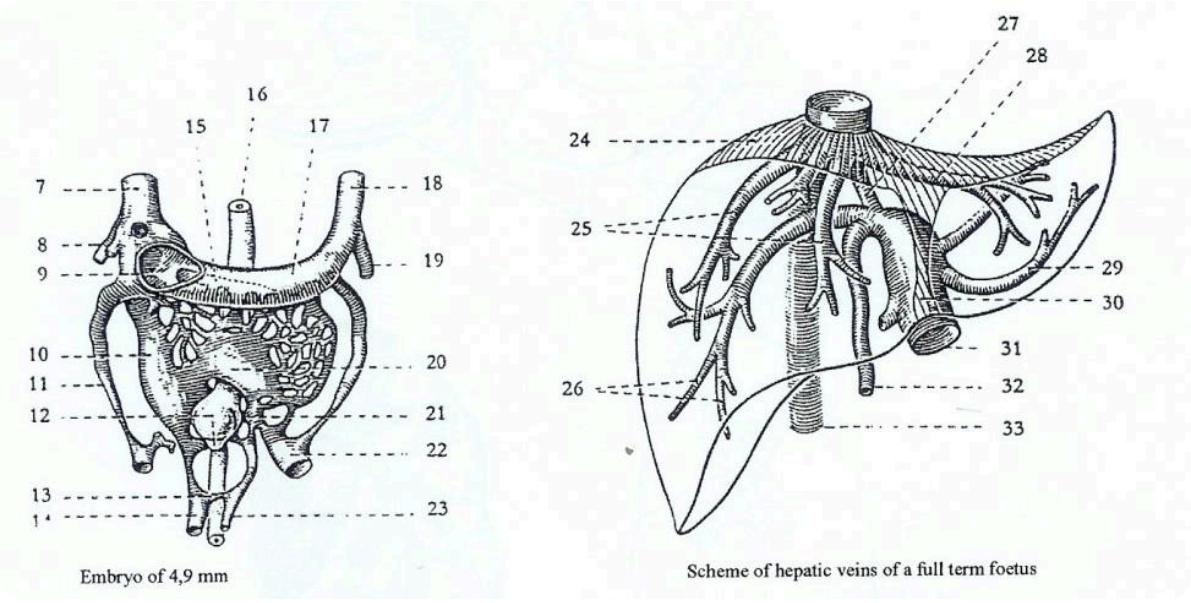


FIG. — Embryo, hepatic veins

THE TWO CURRENTS OF THE HEPATIC PORTAL SYSTEM

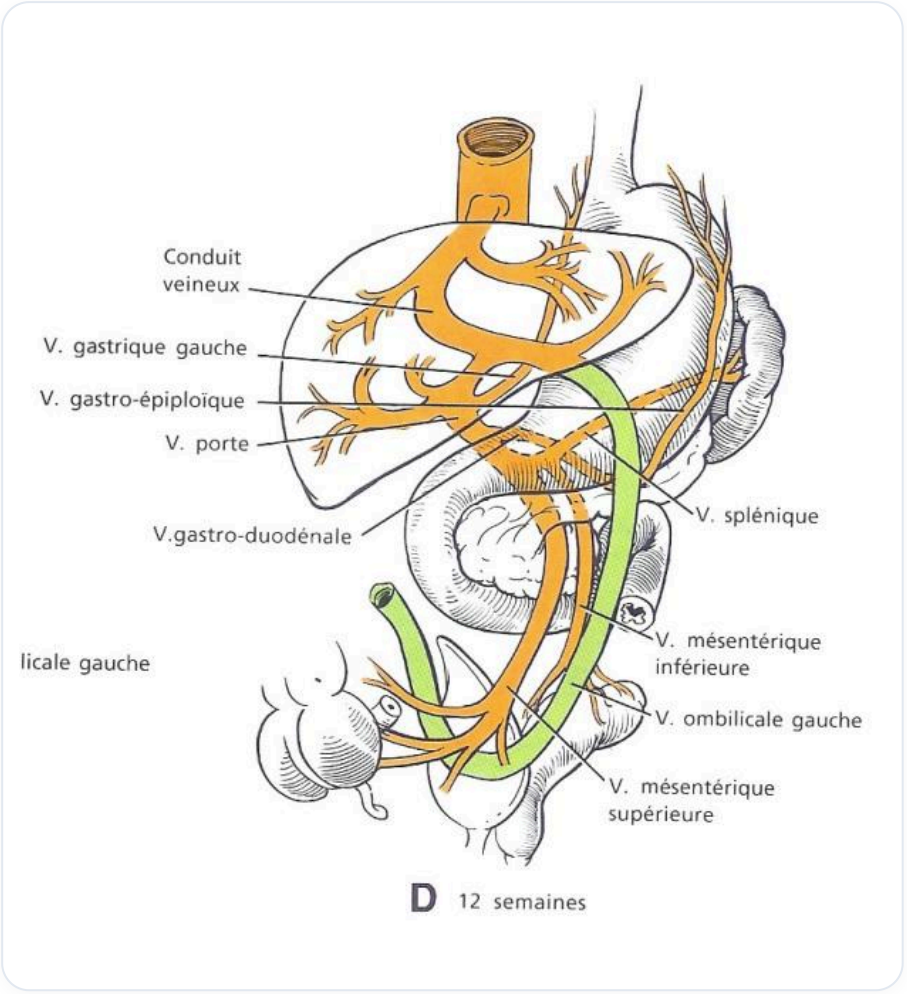


FIG. — Portal system 12 weeks

All blood coming from the **right mesenteric** side is drained by the portal system and is found mainly in the right liver. In contrast, everything inferior (stomach, spleen, and lower colon) drains more to the left liver.

Two currents can be observed within the portal system itself:

- One current oriented towards the left side.
- Another towards the right side.

The liver plays a digestive and metabolic role, but also a motor and emotional one. Organs like the stomach and spleen are related to the emotional system, dreams, and nightmares, leading to congestion.

The portal trunk is common here, but it is interesting to note that there is a flow that does not mix or can mix.

LIVER REGENERATION

The liver never regenerates at its endodermal level, but always regenerates at its mesodermal level. It can rebuild vessels, but not ducts.

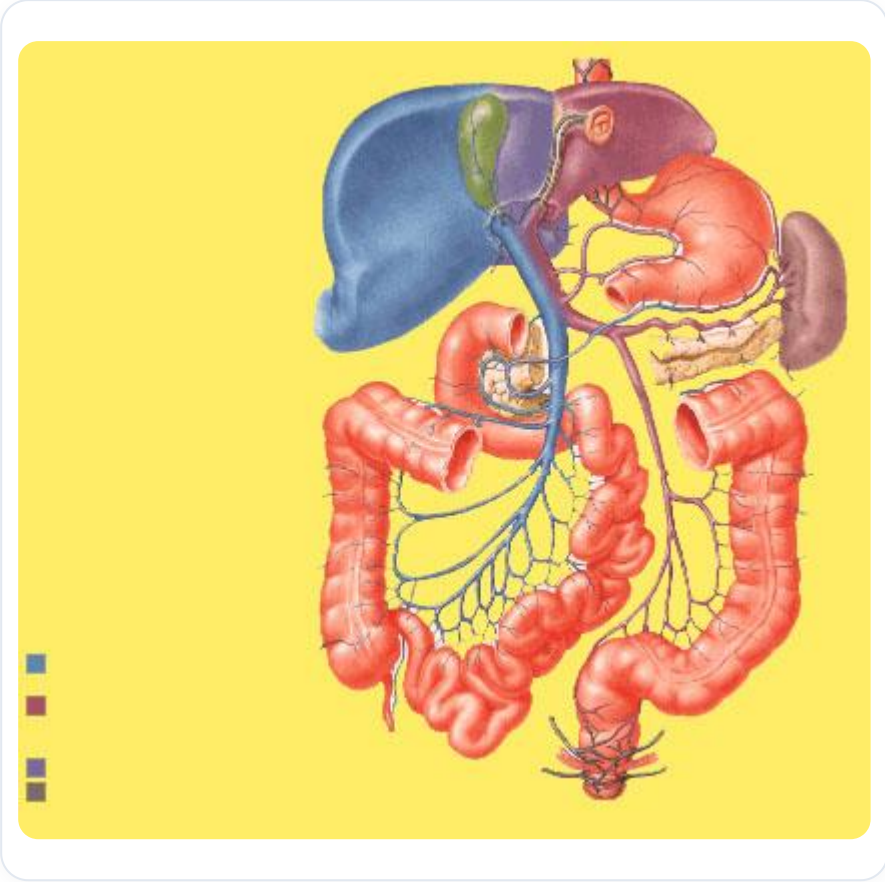


FIG. — Hepatic portal system

The liver is a set of **bile ducts** integrated with many vessels. It is important to remember that the liver only regenerates at night. There are three levels:

1. **When you eat:** A lot of venous blood and little arterial blood. The liver does not regenerate.

2. **Between meals:** Balance between arterial blood and venous blood. The liver regenerates a little.

3. **At night, when you sleep:** A lot of arterial blood and little venous blood.

Fasting is beneficial because it allows the liver to regenerate and cells to become stronger, releasing energy from the inside out.

THE HEART-LIVER-SPLEEN-SMALL INTESTINE FUNCTIONAL UNIT

A rebalancing occurs between the heart, liver, spleen, and mesenteric system (small intestine). This constitutes a functional unit.

The liver is a volume regulator, capable of increasing or decreasing **blood volume**. It plays a crucial role in embryonic development and at birth.

Bone and tissue compressions, as well as vascular changes, are observed. It is crucial to treat these aspects at a fluid level at birth, in addition to the compressive level.

IMPORTANCE OF CONFLUENCES

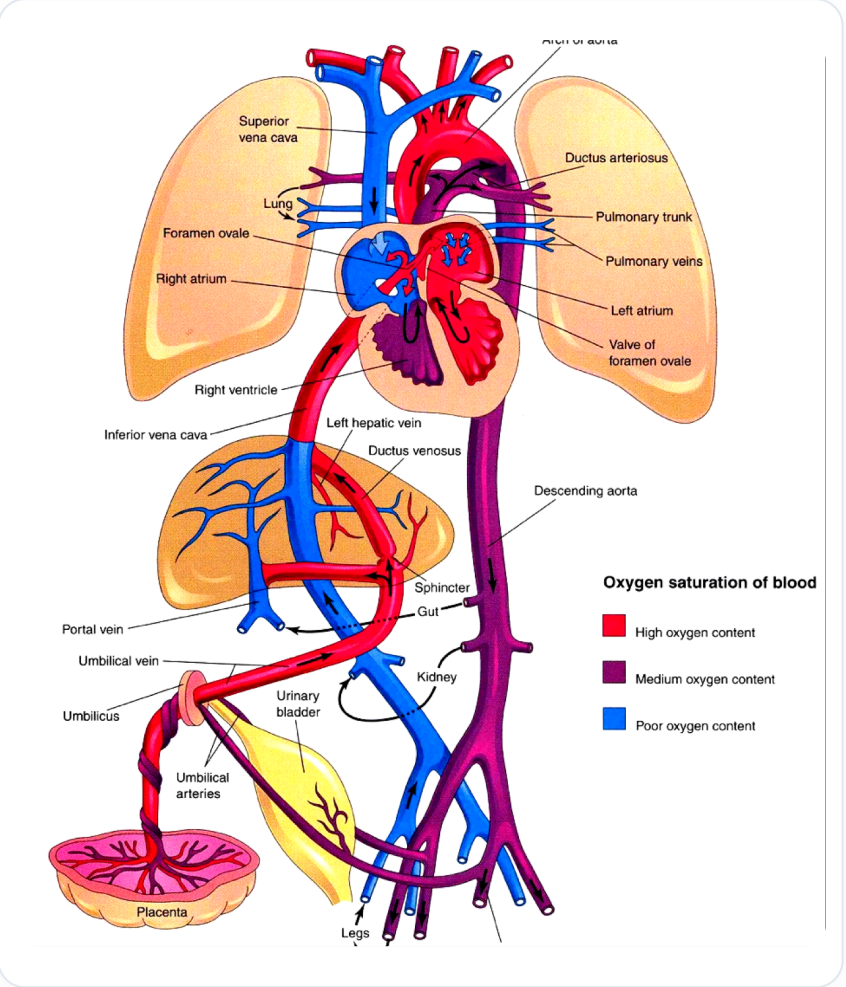


FIG. — Fetal circulation

Retrohepatic confluence is important for heart return. The **subhepatic confluence** and the **jugular confluence** represent two major keys.

It is essential not to overload the heart in terms of venous return. The heart must function correctly in systole, not in diastole. If the liver is systolic, it helps to resorb.

It is therefore necessary to release facial function to restore hepatic release. Do not forget the heart-liver-spleen connection.

VITELLINE VESICLE AND UMBILICAL CORD CONNECTION

The movement of the amniotic cavity shows the vitelline vesicle regressing and depositing on the umbilical cord. A loop will develop, integrating the intestinal loop inside.

In the umbilical cord, the spiralling vessels in the embryonic pedicle connect, with the remnant of the **yolk sac**. The intestinal tube and the embryonic pedicle join, integrating the allantois and the entire process with the large amniotic cavity.

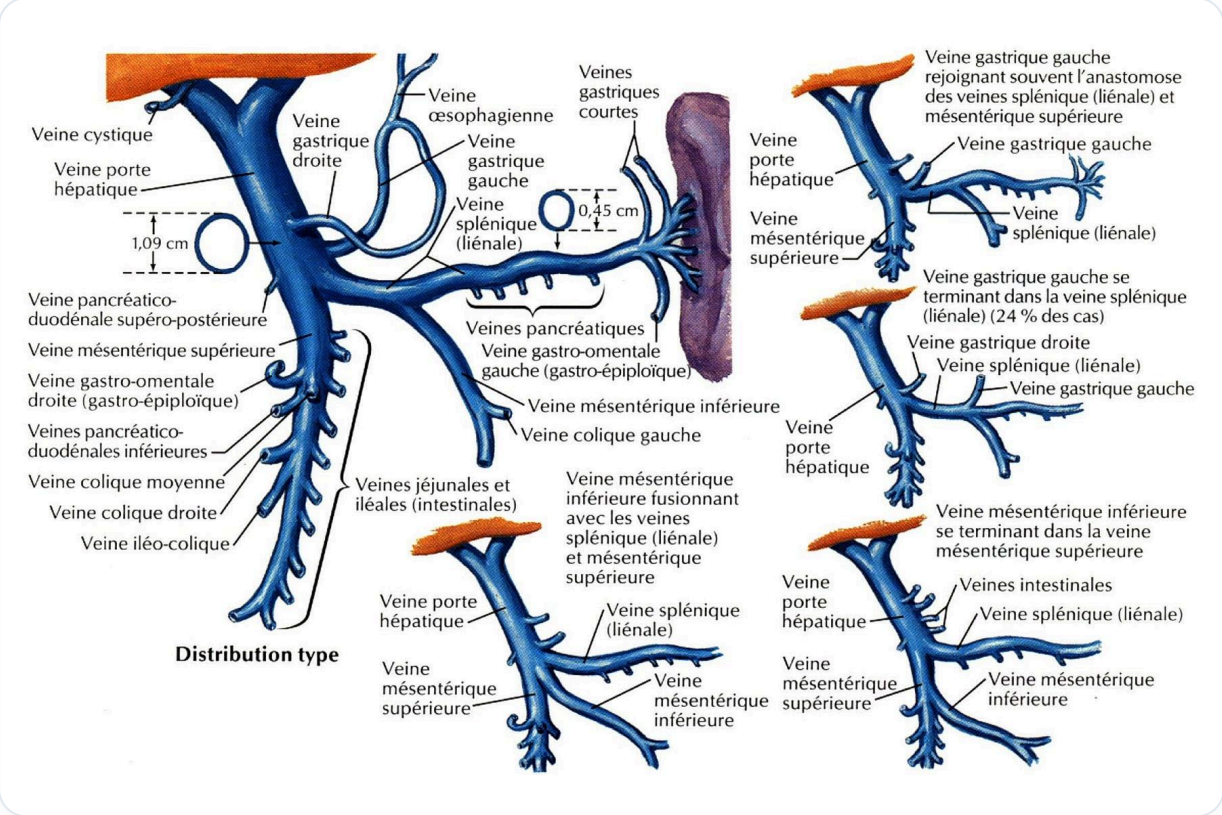


FIG. — Portal Vein Anatomy

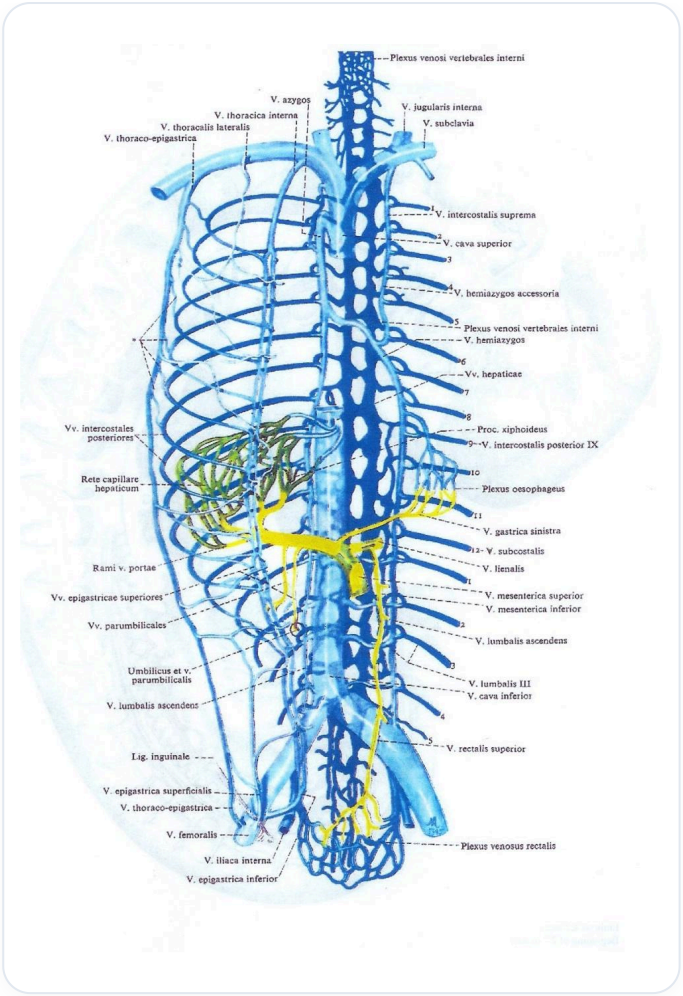


FIG. — Human venous system

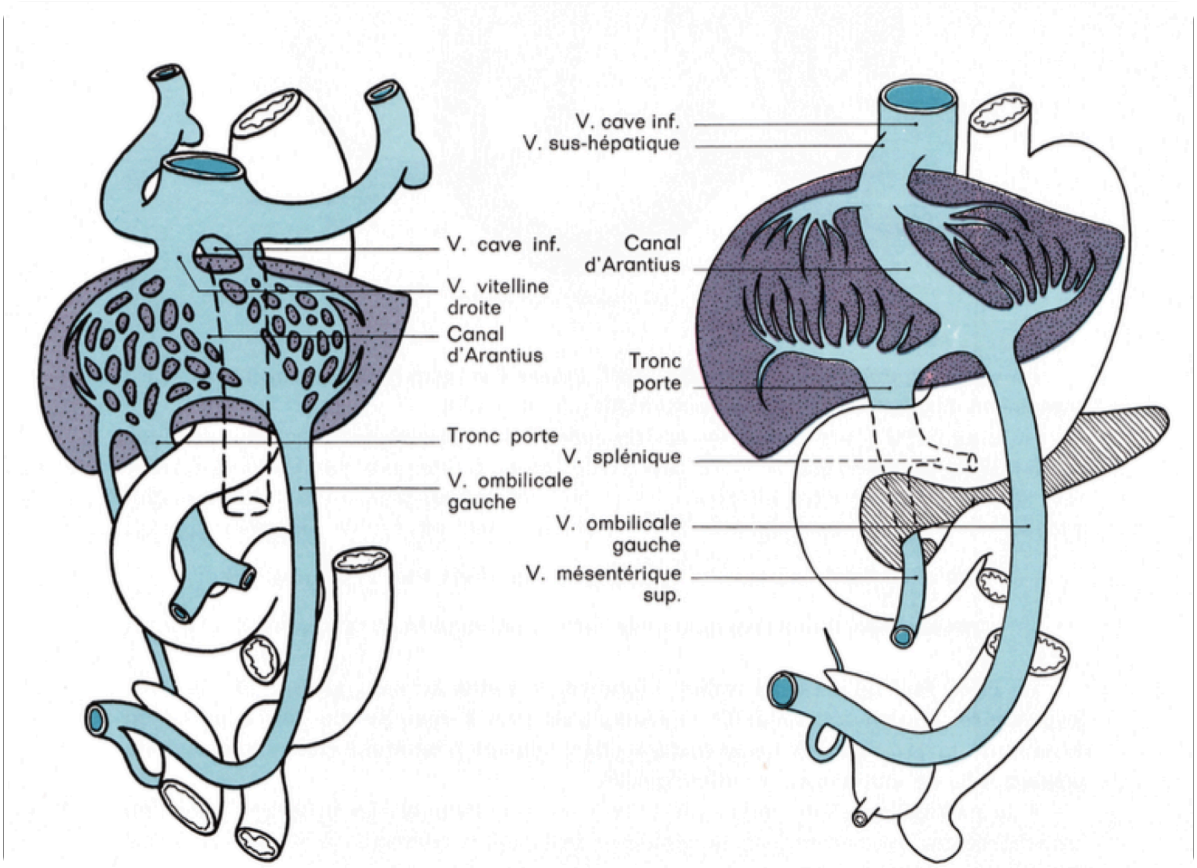


FIG. — Embryonic portal system

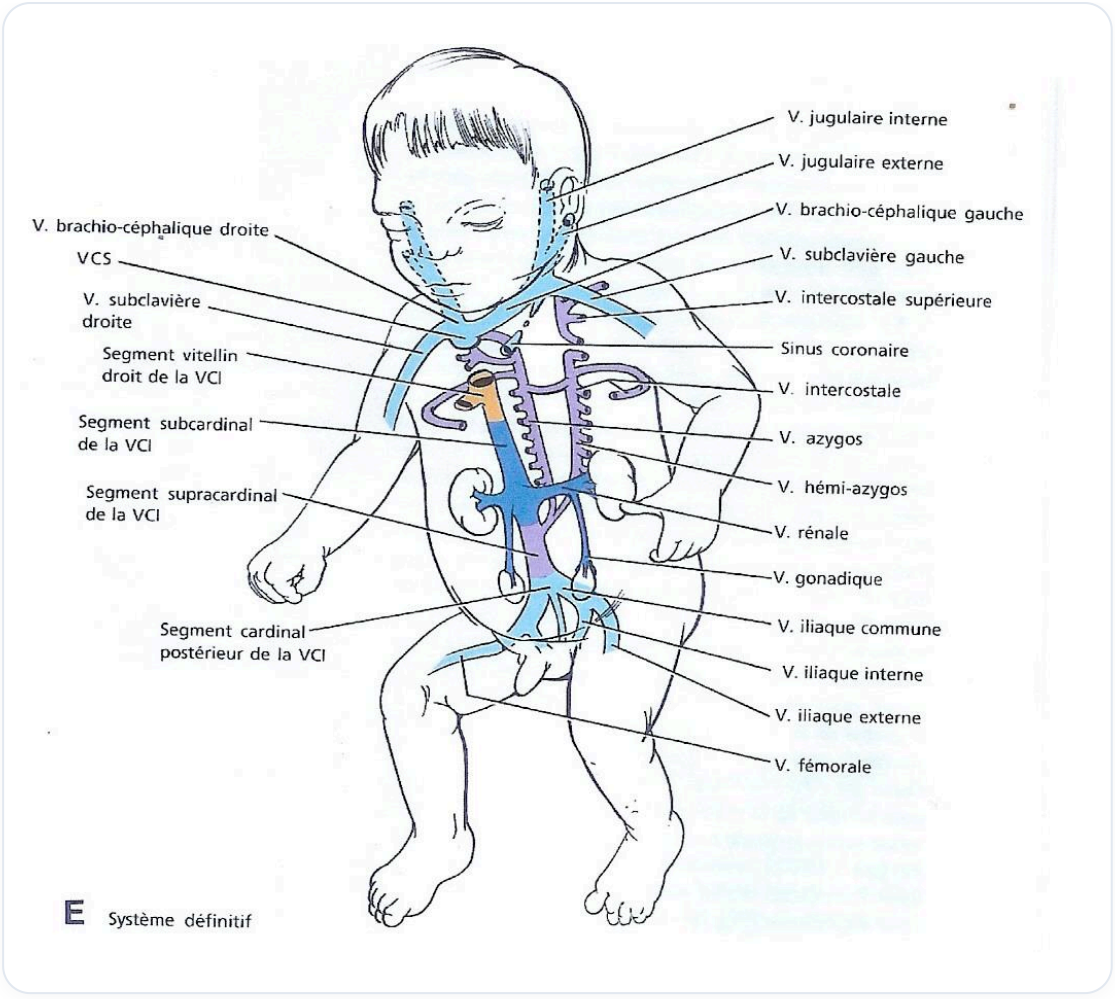


FIG. — Definitive venous system